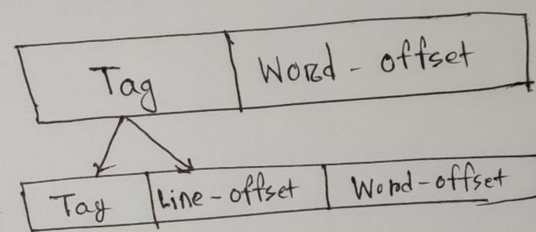
cache mapping:

There are three difference types of mapping

- Direct mapping
- Associative mapping
- Set-Associative mapping

Direct mapping:

Main memory



cache memory

* Differcence between Hardware and softwore:-

Hardware	Softwere
Hardware is a physical part of the computer	software is a set of instuction that tells a computer exactly what to do
Hardware canbe ~~tau~~ touched	softwere but cannot touch them.
Hardware cannotbe infected by virus	The softwere canbe infected by virus
Hardwere cannot perform any task without softwere	The softwere cannotbe executed without hardwere.
It is manufactured	It is developed and engineered
Electronic and other materials are use to create hardwere	created by utilizing a computer language to write instruction.
Ex: keyboard, mouse, Monitor etc	Ex: ms word, Excel, powerpoint

Computer Architecture and Computer Organisation :

Computer Architecture	Computer Organisation
Computer Architecture define the logical aspect of a computer.	Computer organisation defines the physical aspects of the computer system.
Computer Architecture describes what the computer does.	The computer organization describe how it does it.
Computer Architecture deals with the functional behavior.	Computer Organisation deals with a ~~st~~ structural relationship.
It's deals with high-level design issues.	It's deals with Low level desing issues.
Computer Architecture comprises logical fuctions such as instruetion sets, registers, data types.	Computer organization consists of physical units like circuit desings, peripherals and adders.

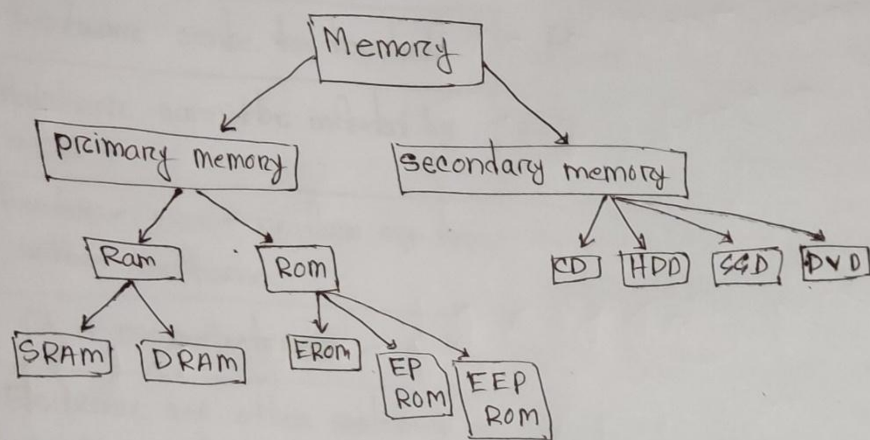
Que : Why is the CPU referred to as the brain of a computer?

Ans : Memory computer Architeach slide page 2.

Memory:

= Memory is the devices the sotores and retrieves data.

* categories of memory



primary memory: primary memory is the memory that the cpu can access directly.

secondary memory: Secondery memory is the memory that the cpu cannot access directly.

* Differenence between primary memory and secondry memory.

Primary memory	Secondary memory
• Main mamory	External memory
• Store data temporarly	• Save data permanently
• Faster	• Slower
• Smaller capacity	• Larger capacity
• Primary memory is directly accessible by cpu	• Secondary memory is not directly accessible by the cpu
Ex: RAM, Rom, PROM	• Ex: Hard Disk, Floppy Disk

RAM: (Random access memory) is a type of volatile computer memory that temporarily sotores data

Rom: (Read only memiry) is a type of non-volatile ~~Computer~~ memory used in computers and electronic devices to permanently store data.

Difference between Ram and Rom:

RAM	Rom
• Random access memory	• Read only memory
• volatile	• Non volatile
• Support read and write	• Support only read operation
• It is a high speed memory	• It is a much slower memory
• Memory capacity Larger	• Memory capacity smaller
• CPU can easily access data stored in RAM	• CPU cannot easily access data stored in Rom.

Difference between HDD and SSD

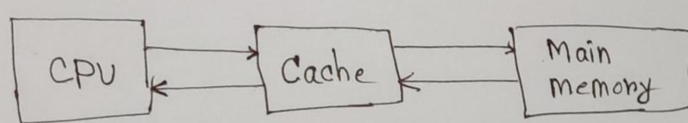
HDD	SSD
• Hard Disk Drive	• Solid state Drive
• High storage capacity	• Lower storage capacity
• Lower cost	• Higher cost compare to HDD
• Slower performance	• ~~First~~ Fast compare to SSD
• HDD have moving parts	• SSD have no moving parts
• HDDs store data in ~~meg~~ magnetic disks	• SSDs store data in flash memory

* Compare different memory:

- ~~Re Resisten~~
- Register
- SRAM
- DRAM
- SSD
- HDD

* ~~case~~ cache memory:

cache memory is a small storage component located close to the cpu that temponarily store freequently accessed data and instruction.



Charcacteristics of cache memory –

- Cache memory is a fast memory
- cache memory holds freequently data and instuctions.
- cache memory is costlier than main memory.
- Cache memory is used to speed up and synchronize with a high speed cpu.

* Cache performance:
  - Hit
  - miss

Miss and Hit formula:

Hit ratio =